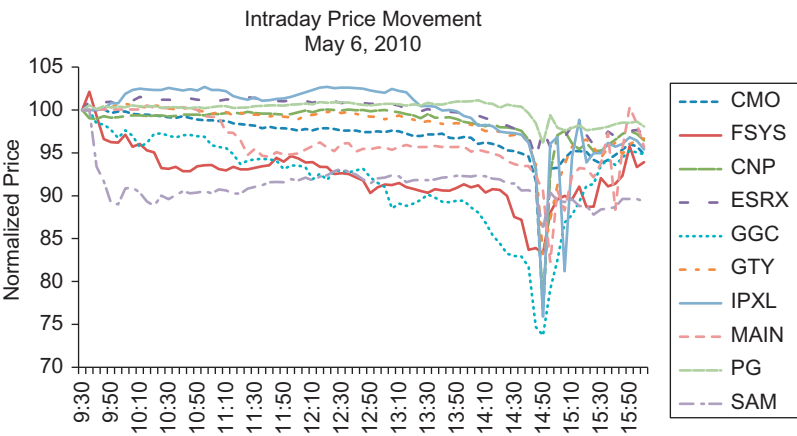


Table 2.12 Sample of Actual Volumes and Price Movement (Flash Crash)

Symbol	Volume Ratio	Percentage Change			Z-Score		
		Close-Close	Open-Close	High-Low	Close-Close	Open-Close	High-Low
IPXL	2.6	−4.4%	−4.4%	−62.1%	−2.2	−2.2	−30.8
SAM	5.6	−0.7%	−6.1%	−52.5%	−0.4	−3.6	−31.0
PG	2.6	−2.3%	−1.9%	−37.2%	−2.9	−2.4	−46.9
GTY	2.9	−3.9%	−3.5%	−35.9%	−2.5	−2.2	−22.6
MAIN	6.2	−4.7%	−4.3%	−33.0%	−2.8	−2.5	−19.7
CNP	3.1	−3.5%	−3.5%	−31.4%	−2.7	−2.8	−24.4
ESRX	2.7	−2.8%	−2.9%	−27.6%	−2.0	−2.0	−19.2
GGC	3.1	−2.4%	−5.7%	−27.5%	−0.8	−1.9	−9.4
CMO	2.7	−4.9%	−4.6%	−25.3%	−2.7	−2.6	−14.0
FSYS	4.3	−4.3%	−5.8%	−20.4%	−1.6	−2.2	−7.6
Avg	3.6	−3.4%	−4.3%	−35.3%	−2.1	−2.4	−22.6
SPY	3.3	−3.4%	−2.9%	−10.8%	−3.5	−3.0	−11.3



■ **Figure 2.9** Intraday Price Movement (Sample of Stock).

this magnitude to occur about 5 times a year. So this type of movement is not that out of the ordinary or unexpected. However, the average high-low price range was -35.3% and is equivalent to a -22.6 sigma event! We have heard of 6 sigma before, well this is almost 4 times higher than a 6 sigma event! This magnitude decline in prices followed by a price reversal of almost the same magnitude is highly unlikely and unexpected. Volumes